

Algebraic Geometry 7

Lecture notes with examples

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Contents

1 Theory for Problems 8–10: Normalization, Fibres, and Cartesian Diagrams

This section collects the main theoretical ideas behind Problems 8–10. The three problems are connected by one common theme:

geometric constructions are controlled by local algebra and universal properties.

Problem 8 concerns normalization, which is built from integral closure and gluing. Problem 9 concerns fibres of morphisms, which are computed by base change to residue fields. Problem 10 concerns Cartesian diagrams, which are diagrams characterized by the universal property of fibre products.

1.1 Normal Schemes and Normalization

Integral schemes

A scheme X is called **integral** if it is both reduced and irreducible.

Equivalently, X is integral if every nonempty affine open subset

$$U = \operatorname{Spec} A$$

has A an integral domain.

For an integral scheme X , one can define its function field. If

$$U = \operatorname{Spec} A$$

is a nonempty affine open subset, then

$$K(X) = \operatorname{Frac}(A).$$

This does not depend on the chosen affine open subset, because all nonempty affine opens of an integral scheme have compatible fraction fields.

Normal rings

Let A be an integral domain, and let

$$K = \operatorname{Frac}(A)$$

be its field of fractions. The ring A is called **normal** if it is integrally closed in K .

This means that whenever

$$x \in K$$

satisfies a monic polynomial equation

$$x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0, \quad a_i \in A,$$

then

$$x \in A.$$

Thus a normal domain contains all elements of its fraction field that are forced algebraically by A .

Integral closure

Let A be a domain with fraction field K . The **integral closure** of A in K is

$$\overline{A} = \{x \in K : x \text{ is integral over } A\}.$$

It satisfies

$$A \subseteq \overline{A} \subseteq K.$$

The ring $\overline{A}$ is integrally closed in K . Therefore $\overline{A}$ is the smallest normal domain inside K containing A .

If A is already normal, then

$$\overline{A} = A.$$

If A is not normal, then $\overline{A}$ is a canonical enlargement of A obtained by adding all integral elements from the fraction field.

Normal schemes

A scheme X is called **normal** if all local rings

$$\mathcal{O}_{X,x}$$

are integrally closed domains.

For an integral scheme, normality can be checked affine-locally. Thus X is normal if and only if for every affine open subset

$$U = \operatorname{Spec} A \subseteq X,$$

the ring A is a normal domain.

So normality is a local property.

Normalization of an affine integral scheme

If

$$X = \operatorname{Spec} A$$

is an affine integral scheme, then its normalization is

$$\widetilde{X} = \operatorname{Spec} \overline{A},$$

where $\overline{A}$ is the integral closure of A in its fraction field

$$K = \operatorname{Frac}(A).$$

The inclusion

$$A \hookrightarrow \overline{A}$$

induces a morphism

$$\pi : \widetilde{X} \rightarrow X.$$

This is the affine normalization morphism.

Normalization and localization

The key algebraic fact behind normalization is that integral closure commutes with localization.

Let $S \subseteq A$ be a multiplicative subset. Then

$$\overline{S^{-1}A} = S^{-1}\overline{A}.$$

In particular, for a distinguished open subset

$$D(f) = \operatorname{Spec} A_f \subseteq \operatorname{Spec} A,$$

we have

$$\overline{A_f} = (\overline{A})_f.$$

Therefore the normalization of $D(f)$ is

$$\mathrm{Spec} \overline{A_f} = \mathrm{Spec}(\overline{A})_f.$$

But this is exactly the open subset of

$$\mathrm{Spec} \overline{A}$$

lying above $D(f)$.

This compatibility is the reason that affine normalizations glue.

Normalization of an integral scheme

Let X be an integral scheme. Choose an affine open cover

$$X = \bigcup_i U_i, \quad U_i = \mathrm{Spec} A_i.$$

For each affine open U_i , define

$$\widetilde{U}_i = \mathrm{Spec} \overline{A_i}.$$

Because normalization is compatible with localization, the schemes $\widetilde{U}_i$ agree on overlaps. Therefore they glue to a scheme

$$\widetilde{X}.$$

The local morphisms

$$\mathrm{Spec} \overline{A_i} \rightarrow \mathrm{Spec} A_i$$

also glue to a global morphism

$$\pi : \widetilde{X} \rightarrow X.$$

The scheme $\widetilde{X}$ is called the **normalization** of X .

Universal property of normalization

The normalization has the following universal property.

Let Z be a normal integral scheme, and let

$$f : Z \rightarrow X$$

be a dominant morphism. Then there exists a unique morphism

$$\tilde{f} : Z \rightarrow \tilde{X}$$

such that

$$f = \pi \circ \tilde{f}.$$

Equivalently, the following diagram commutes:

$$\begin{array}{ccc} Z & \xrightarrow{\tilde{f}} & \tilde{X} \\ & \searrow f & \swarrow \pi \\ & X & \end{array}$$

The reason is as follows. Since f is dominant and X, Z are integral, f induces an inclusion of function fields

$$K(X) \hookrightarrow K(Z).$$

If $U = \operatorname{Spec} A \subseteq X$ is affine and $W = \operatorname{Spec} B \subseteq f^{-1}(U)$ is affine, then the morphism gives a ring map

$$A \rightarrow B.$$

Every element of $\bar{A}$ is integral over A . Its image in $K(Z)$ is therefore integral over B . Since Z is normal, B is integrally closed. Hence the image actually lies in B .

Thus the map

$$A \rightarrow B$$

extends uniquely to

$$\bar{A} \rightarrow B.$$

These local extensions glue and give the unique factorization

$$Z \rightarrow \tilde{X}.$$

Thus normalization is the universal normal integral scheme mapping birationally to X .

1.2 Fibres of Morphisms of Affine Schemes

The general fibre formula

Let

$$\varphi : A \rightarrow B$$

be a ring homomorphism. It induces a morphism of affine schemes

$$f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A.$$

Let

$$\mathfrak{p} \in \operatorname{Spec} A$$

be a point. The residue field of $\mathfrak{p}$ is

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}).$$

The scheme-theoretic fibre of f over $\mathfrak{p}$ is

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} (B \otimes_A \kappa(\mathfrak{p})).$$

Thus the fundamental formula is

$$\boxed{f^{-1}(\mathfrak{p}) = \operatorname{Spec} (B \otimes_A \kappa(\mathfrak{p}))}.$$

Geometric meaning of the fibre formula

The formula

$$B \otimes_A \kappa(\mathfrak{p})$$

means that we specialize the base ring A to the residue field at the point $\mathfrak{p}$.

For example, suppose

$$B = k[T, U]/(F(T, U))$$

and the morphism is

$$\operatorname{Spec} B \rightarrow \operatorname{Spec} k[T].$$

If the point of the base corresponds to the value $T = a$, then the fibre is morally obtained by substituting

$$T = a$$

into the equation $F(T, U)$. Therefore the fibre is

$$\operatorname{Spec} k[U]/(F(a, U))$$

when the residue field is k . More generally, it is

$$\operatorname{Spec} \kappa(\mathfrak{p})[U]/(F(a, U)),$$

where

$$a = \overline{T} \in \kappa(\mathfrak{p}).$$

Generic points and closed points

For the affine line

$$\operatorname{Spec} k[T],$$

there is a generic point

$$(0),$$

with residue field

$$\kappa((0)) = k(T).$$

The closed points are ideals

$$(p(T)),$$

where $p(T) \in k[T]$ is irreducible. Their residue fields are

$$\kappa((p(T))) = k[T]/(p(T)).$$

If k is algebraically closed, then every closed point has the form

$$(T - a), \quad a \in k,$$

and its residue field is k .

If k is not algebraically closed, closed points may have larger residue fields.

Fibres over $\operatorname{Spec} \mathbb{Z}$

The scheme

$$\operatorname{Spec} \mathbb{Z}$$

has one generic point

$$(0)$$

and one closed point

$$(p)$$

for each prime number p .

The residue field at the generic point is

$$\kappa((0)) = \mathbb{Q}.$$

The residue field at the closed point (p) is

$$\kappa((p)) = \mathbb{F}_p.$$

Therefore, if

$$X \rightarrow \operatorname{Spec} \mathbb{Z}$$

is a scheme over $\mathbb{Z}$, then its generic fibre is

$$X_{\mathbb{Q}} = X \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Q},$$

and its fibre over p is

$$X_{\mathbb{F}_p} = X \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{F}_p.$$

So a scheme over $\operatorname{Spec} \mathbb{Z}$ is like a family whose fibres live in different characteristics.

Reduced fibres

A scheme

$$\operatorname{Spec} R$$

is reduced if the ring R has no nonzero nilpotent elements.

That means

$$x^n = 0 \implies x = 0.$$

For example,

$$k[U]/(U^2)$$

is not reduced, because the class of U is nonzero but satisfies

$$U^2 = 0.$$

Therefore

$$\operatorname{Spec} k[U]/(U^2)$$

is a nonreduced scheme. Geometrically, it is a doubled point: it has one underlying topological point, but with infinitesimal thickness.

Irreducible fibres

A topological space is irreducible if it cannot be written as the union of two proper closed subsets.

For affine schemes, $\operatorname{Spec} R$ is irreducible if and only if the nilradical

$$\sqrt{(0)}$$

is a prime ideal.

In particular:

$$R \text{ is a domain} \implies \operatorname{Spec} R \text{ is irreducible.}$$

However, irreducibility and reducedness are different properties.

For example:

$$\operatorname{Spec} k[U]/(U^2)$$

is irreducible, because its underlying topological space has only one point. But it is not reduced, because U is nilpotent.

On the other hand,

$$\operatorname{Spec}(k \times k)$$

is reduced but not irreducible, because it consists of two separate reduced points.

Common fibre types

The following table summarizes common fibres appearing in Problem 9:

Ring	Geometric fibre	Reduced?	Irreducible?
k	one point	yes	yes
$k \times k$	two points	yes	no
$k[U]/(U^2)$	doubled point	no	yes
0	empty fibre	yes	usually no

This table is useful when analysing fibres after tensoring with residue fields.

Splitting, ramification, and inertia

The morphism

$$\operatorname{Spec} \mathbb{Z}[T]/(T^2 + 1) \rightarrow \operatorname{Spec} \mathbb{Z}$$

is the same as

$$\operatorname{Spec} \mathbb{Z}[i] \rightarrow \operatorname{Spec} \mathbb{Z}.$$

The fibre over a prime p is

$$\operatorname{Spec} \mathbb{F}_p[T]/(T^2 + 1).$$

Thus the geometry of the fibre depends on the factorization of

$$T^2 + 1$$

modulo p .

If $p = 2$, then

$$T^2 + 1 = (T + 1)^2$$

in $\mathbb{F}_2[T]$. Hence the fibre is a nonreduced doubled point. Number-theoretically, 2 is ramified in $\mathbb{Z}[i]$.

If

$$p \equiv 1 \pmod{4},$$

then -1 is a square modulo p , so $T^2 + 1$ splits into two distinct linear factors. The fibre consists of two reduced points. Number-theoretically, p splits in $\mathbb{Z}[i]$.

If

$$p \equiv 3 \pmod{4},$$

then -1 is not a square modulo p , so $T^2 + 1$ remains irreducible. The fibre is

$$\mathrm{Spec} \mathbb{F}_{p^2}.$$

Number-theoretically, p is inert in $\mathbb{Z}[i]$.

Thus:

Prime p	Fibre	Number-theoretic meaning
2	nonreduced doubled point	ramified
$p \equiv 1 \pmod{4}$	two reduced points	split
$p \equiv 3 \pmod{4}$	$\mathrm{Spec} \mathbb{F}_{p^2}$	inert

This example shows how scheme-theoretic fibres encode classical arithmetic information.

1.3 Fibre Products and Cartesian Diagrams

Fibre products

Let

$$X \rightarrow S$$

and

$$Y \rightarrow S$$

be morphisms in a category. Their fibre product is an object

$$X \times_S Y$$

together with projection maps

$$X \times_S Y \rightarrow X, \quad X \times_S Y \rightarrow Y,$$

such that the two compositions to S are equal.

The diagram is

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & S \end{array}$$

The universal property says that for every object Q , giving a morphism

$$Q \rightarrow X \times_S Y$$

is equivalent to giving morphisms

$$Q \rightarrow X, \quad Q \rightarrow Y,$$

such that the two compositions to S agree.

In symbols,

$$\mathrm{Hom}(Q, X \times_S Y) \cong \mathrm{Hom}(Q, X) \times_{\mathrm{Hom}(Q, S)} \mathrm{Hom}(Q, Y).$$

This is the most important tool for proving that a diagram is Cartesian.

Cartesian diagrams

A commutative square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

is called **Cartesian** if the natural morphism

$$A \rightarrow B \times_D C$$

is an isomorphism.

Equivalently, A satisfies the same universal property as the fibre product $B \times_D C$.

Thus, for every object Q , giving a morphism

$$Q \rightarrow A$$

is equivalent to giving morphisms

$$Q \rightarrow B, \quad Q \rightarrow C,$$

such that their compositions to D agree.

Base change

A Cartesian square can also be understood as a **base change square**.

Suppose

$$X \rightarrow S$$

is a morphism, and suppose

$$S' \rightarrow S$$

is another morphism. Then the fibre product

$$X_{S'} = X \times_S S'$$

is called the base change of X from S to S' .

The diagram

$$\begin{array}{ccc} X \times_S S' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

is Cartesian.

Fibres are special cases of base change. The fibre of

$$X \rightarrow S$$

over a point $s \in S$ is

$$X_s = X \times_S \operatorname{Spec} \kappa(s).$$

Thus Problem 9 is also an application of fibre products.

Diagonal morphisms

If

$$X \rightarrow S$$

is a morphism, the **diagonal morphism** is

$$\Delta_X : X \rightarrow X \times_S X.$$

It is induced by the pair

$$(\operatorname{id}_X, \operatorname{id}_X).$$

Geometrically, it sends a point x to

$$(x, x).$$

The diagonal imposes equality. Pulling back along a diagonal means forcing two maps to become equal.

This idea is central in Problem 10.

Graph morphisms

Let

$$f : X \rightarrow Y$$

be a morphism over S . The **graph morphism** of f is

$$\Gamma_f : X \rightarrow X \times_S Y.$$

It is defined by

$$\Gamma_f = (\text{id}_X, f).$$

Thus a point $x \in X$ is sent to

$$(x, f(x)).$$

The graph remembers both the point and its image under f .

First Cartesian diagram in Problem 10

Suppose we have morphisms

$$X \rightarrow S \rightarrow T, \quad Y \rightarrow S \rightarrow T.$$

Then we have two fibre products:

$$X \times_S Y$$

and

$$X \times_T Y.$$

The condition defining $X \times_T Y$ is weaker: it only says that the images of X and Y agree after mapping to T .

The condition defining $X \times_S Y$ is stronger: it says that the images of X and Y already agree after mapping to S .

The diagonal

$$\Delta : S \rightarrow S \times_T S$$

imposes precisely the condition that two maps to S are equal.

Therefore

$$X \times_S Y$$

can be recovered from

$$X \times_T Y$$

by pulling back along the diagonal

$$S \rightarrow S \times_T S.$$

That is,

$$X \times_S Y \cong (X \times_T Y) \times_{S \times_T S} S.$$

This is exactly the statement that the diagram

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & S \\ \downarrow & & \downarrow \Delta \\ X \times_T Y & \longrightarrow & S \times_T S \end{array}$$

is Cartesian.

Second Cartesian diagram in Problem 10

Let

$$f : X \rightarrow Y$$

be a morphism over S . The graph of f is

$$\Gamma_f : X \rightarrow X \times_S Y.$$

The diagonal of Y is

$$\Delta : Y \rightarrow Y \times_S Y.$$

The diagram is

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \Gamma_f \downarrow & & \downarrow \Delta \\ X \times_S Y & \longrightarrow & Y \times_S Y \end{array}$$

The bottom morphism sends

$$(x, y)$$

to

$$(f(x), y).$$

Pulling back along the diagonal imposes the condition

$$(f(x), y) \in \Delta(Y).$$

But this means

$$f(x) = y.$$

Therefore the pullback consists exactly of pairs

$$(x, y)$$

such that

$$y = f(x).$$

This is precisely the graph of f . Hence

$$X \cong (X \times_S Y) \times_{Y \times_S Y} Y.$$

Therefore the square is Cartesian.

1.4 Conceptual Summary

The three problems can be organized as follows:

Problem	Main object	Main idea
8	Normalization	integral closure and gluing
9	Fibres	base change to residue fields
10	Cartesian diagrams	universal property of fibre products

In Problem 8, one starts with an integral scheme X and constructs a normal integral scheme

$$\widetilde{X}$$

together with a morphism

$$\pi : \widetilde{X} \rightarrow X.$$

This construction is local on affine open subsets and is controlled by integral closure.

In Problem 9, one studies a morphism

$$X \rightarrow Y$$

by examining its fibres

$$X_y = X \times_Y \operatorname{Spec} \kappa(y).$$

This reduces geometric questions about fibres to algebraic questions about tensor products and residue fields.

In Problem 10, one proves that certain diagrams are Cartesian by showing that they satisfy the universal property of fibre products.

Thus the common language is

localization, base change, and universal properties.

Normalization is built by localization and gluing. Fibres are obtained by base change to residue fields. Cartesian diagrams are recognized by universal properties.

2 Examples for Problems 8–10: Normalization, Fibres, and Cartesian Diagrams

This section gives examples illustrating the three main constructions from Problems 8–10:

normalization, scheme-theoretic fibres, Cartesian diagrams.

The guiding idea is that these constructions are all controlled by local algebra and universal properties.

2.1 Examples of Normalization

Example 1. The affine line is already normal

Let

$$X = \operatorname{Spec} k[t].$$

The ring $k[t]$ is a principal ideal domain, hence a unique factorization domain, hence integrally closed. Therefore $k[t]$ is normal.

Thus the normalization of X is just X itself:

$$\widetilde{X} = X.$$

The normalization morphism is the identity morphism

$$\mathrm{id}_X : X \rightarrow X.$$

So smooth affine space is already normal.

Example 2. Normalization of the cusp

Consider the cuspidal affine curve

$$X = \mathrm{Spec} k[x, y]/(y^2 - x^3).$$

Let

$$A = k[x, y]/(y^2 - x^3).$$

The curve has a parametrization

$$x = t^2, \quad y = t^3.$$

This gives an embedding

$$A \hookrightarrow k[t]$$

defined by

$$x \mapsto t^2, \quad y \mapsto t^3.$$

Hence

$$A \cong k[t^2, t^3] \subseteq k[t].$$

The fraction field of A is

$$\mathrm{Frac}(A) = k(t).$$

The element $t \in k(t)$ is integral over A , because it satisfies the monic equation

$$T^2 - x = 0.$$

Indeed,

$$t^2 = x \in A.$$

However,

$$t \notin k[t^2, t^3].$$

Therefore A is not integrally closed. Its integral closure in $k(t)$ is

$$\overline{A} = k[t].$$

Thus the normalization of X is

$$\widetilde{X} = \operatorname{Spec} k[t].$$

The normalization morphism

$$\pi : \operatorname{Spec} k[t] \rightarrow \operatorname{Spec} k[x, y]/(y^2 - x^3)$$

is induced by the ring map

$$k[x, y]/(y^2 - x^3) \rightarrow k[t],$$

where

$$x \mapsto t^2, \quad y \mapsto t^3.$$

Geometrically, the affine line parametrizes the cusp and removes the singularity.

Example 3. The algebraic form $k[t^2, t^3] \subseteq k[t]$

Let

$$A = k[t^2, t^3].$$

Then

$$A \subseteq k[t], \quad \operatorname{Frac}(A) = k(t).$$

The element t is integral over A , since

$$t^2 \in A.$$

Thus t satisfies

$$T^2 - t^2 = 0,$$

where $t^2 \in A$. However,

$$t \notin A.$$

Therefore A is not normal.

The integral closure of A in $k(t)$ is

$$\overline{A} = k[t].$$

Hence

$$\operatorname{Spec} k[t] \rightarrow \operatorname{Spec} k[t^2, t^3]$$

is the normalization.

This is one of the standard examples showing that normalization adds missing integral elements.

Example 4. Normalization separates branches of a node

Consider the nodal-type curve

$$X = \operatorname{Spec} k[x, y]/(y^2 - x^2(x + 1)).$$

The equation is

$$y^2 = x^2(x + 1).$$

At the origin, the curve has a singularity. Locally, the equation behaves like

$$y^2 = x^2,$$

so there are two branches meeting at the origin.

Normalization separates these branches. In contrast with the cusp, where the singularity has one branch, a node has two branches. The normalization map has two points lying over the node, one for each branch.

Thus:

normalization of a cusp smooths one branch,

while

normalization of a node separates two branches.

Example 5. Normalization is not always finite without hypotheses

For many schemes occurring in algebraic geometry, especially Noetherian excellent schemes, normalization is finite.

However, in complete generality, the integral closure

$$\overline{A}$$

of a domain A in its fraction field can be very large. Therefore the normalization morphism

$$\mathrm{Spec} \overline{A} \rightarrow \mathrm{Spec} A$$

need not be finite without additional assumptions.

This is why finiteness of normalization is usually stated under hypotheses such as Noetherianity, excellence, or finite type over a field.

2.2 Examples of Scheme-Theoretic Fibres

Example 6. Projection from the hyperbola

Consider

$$X = \mathrm{Spec} k[T, U]/(TU - 1)$$

with the morphism

$$X \rightarrow \mathrm{Spec} k[T].$$

Geometrically, this is the hyperbola

$$TU = 1$$

projected to the T -axis.

The fibre over $T = a$ is obtained by substituting $T = a$. Algebraically, it is

$$\mathrm{Spec} k[U]/(aU - 1).$$

If

$$a \neq 0,$$

then

$$aU - 1 = 0$$

has the unique solution

$$U = a^{-1}.$$

Hence the fibre is one reduced point:

$$\operatorname{Spec} k.$$

If

$$a = 0,$$

then the equation becomes

$$0 \cdot U = 1,$$

which is impossible. Algebraically,

$$k[U]/(-1) = 0.$$

Hence the fibre is empty:

$$\operatorname{Spec} 0 = \emptyset.$$

Thus

Base point	Fibre	Reduced?	Irreducible?
$T = 0$	$\emptyset$	yes	usually no
$T = a \neq 0$	$\operatorname{Spec} k$	yes	yes

Geometrically, this morphism is the inclusion

$$\mathbb{A}_k^1 \setminus \{0\} \rightarrow \mathbb{A}_k^1.$$

Example 7. A double cover with a ramified fibre

Consider

$$X = \operatorname{Spec} k[T, U]/(U^2 - T)$$

with the morphism

$$X \rightarrow \operatorname{Spec} k[T].$$

The fibre over $T = a$ is

$$\operatorname{Spec} k[U]/(U^2 - a).$$

If $a \neq 0$ and $a = b^2$ is a square in k , then

$$U^2 - a = (U - b)(U + b).$$

If $\text{char } k \neq 2$, these are distinct factors, so the fibre consists of two reduced points.

If $a = 0$, then the fibre is

$$\text{Spec } k[U]/(U^2).$$

This is a doubled point. It is irreducible but not reduced.

Thus the map

$$U \mapsto U^2$$

is a two-to-one cover away from 0, but ramifies at 0.

Base point	Fibre	Reduced?	Irreducible?
$a \neq 0, a = b^2$	two points	yes	no
$a = 0$	$\text{Spec } k[U]/(U^2)$	no	yes

Example 8. A family with a nonreduced special fibre

Consider

$$X = \text{Spec } k[T, U]/(U^2 - T^2)$$

mapping to

$$\text{Spec } k[T].$$

The fibre over $T = a$ is

$$\text{Spec } k[U]/(U^2 - a^2).$$

If $\text{char } k \neq 2$ and $a \neq 0$, then

$$U^2 - a^2 = (U - a)(U + a),$$

so the fibre consists of two reduced points.

But over $a = 0$, the fibre is

$$\text{Spec } k[U]/(U^2).$$

This is a nonreduced doubled point.

Hence the general fibre and the special fibre have different scheme structures:

Base point	Fibre	Reduced?	Irreducible?
$a \neq 0$	two reduced points	yes	no
$a = 0$	doubled point	no	yes

Example 9. Fibres of the affine line over $\operatorname{Spec} \mathbb{Z}$

Consider

$$\mathbb{A}_{\mathbb{Z}}^1 = \operatorname{Spec} \mathbb{Z}[T].$$

It maps naturally to

$$\operatorname{Spec} \mathbb{Z}.$$

The generic point of $\operatorname{Spec} \mathbb{Z}$ is

$$(0),$$

with residue field

$$\kappa((0)) = \mathbb{Q}.$$

The generic fibre is

$$\operatorname{Spec} (\mathbb{Z}[T] \otimes_{\mathbb{Z}} \mathbb{Q}) = \operatorname{Spec} \mathbb{Q}[T].$$

Thus

$$(\mathbb{A}_{\mathbb{Z}}^1)_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^1.$$

For a prime number p , the closed point (p) has residue field

$$\kappa((p)) = \mathbb{F}_p.$$

The fibre over p is

$$\operatorname{Spec} (\mathbb{Z}[T] \otimes_{\mathbb{Z}} \mathbb{F}_p) = \operatorname{Spec} \mathbb{F}_p[T].$$

Thus

$$(\mathbb{A}_{\mathbb{Z}}^1)_{\mathbb{F}_p} = \mathbb{A}_{\mathbb{F}_p}^1.$$

So the single scheme

$$\operatorname{Spec} \mathbb{Z}[T]$$

contains affine lines in all characteristics:

$$\mathbb{A}_{\mathbb{Q}}^1, \quad \mathbb{A}_{\mathbb{F}_2}^1, \quad \mathbb{A}_{\mathbb{F}_3}^1, \quad \mathbb{A}_{\mathbb{F}_5}^1, \quad \dots$$

Example 10. Gaussian integers and arithmetic fibres

Consider

$$X = \operatorname{Spec} \mathbb{Z}[i].$$

Since

$$\mathbb{Z}[i] \cong \mathbb{Z}[T]/(T^2 + 1),$$

we may write

$$X = \operatorname{Spec} \mathbb{Z}[T]/(T^2 + 1).$$

The natural morphism is

$$X \rightarrow \operatorname{Spec} \mathbb{Z}.$$

The fibre over a prime p is

$$X_p = \operatorname{Spec} \left(\mathbb{F}_p[T]/(T^2 + 1) \right).$$

Now the behaviour depends on the factorization of $T^2 + 1$ modulo p .

For $p = 2$, we have

$$T^2 + 1 = (T + 1)^2$$

in $\mathbb{F}_2[T]$. Hence

$$X_2 = \operatorname{Spec} \mathbb{F}_2[T]/((T + 1)^2).$$

This is a nonreduced doubled point. Number-theoretically, 2 ramifies in $\mathbb{Z}[i]$.

If

$$p \equiv 1 \pmod{4},$$

then -1 is a square modulo p . Therefore $T^2 + 1$ splits into two distinct linear factors, and

$$X_p$$

consists of two reduced points. Number-theoretically, p splits in $\mathbb{Z}[i]$.

If

$$p \equiv 3 \pmod{4},$$

then -1 is not a square modulo p . Therefore $T^2 + 1$ remains irreducible over $\mathbb{F}_p$, and

$$\mathbb{F}_p[T]/(T^2 + 1) \cong \mathbb{F}_{p^2}.$$

Hence

$$X_p = \operatorname{Spec} \mathbb{F}_{p^2}.$$

Number-theoretically, p is inert in $\mathbb{Z}[i]$.

Thus:

Prime p	Fibre	Reduced?	Arithmetic meaning
2	$\operatorname{Spec} \mathbb{F}_2[T]/((T+1)^2)$	no	ramified
$p \equiv 1 \pmod{4}$	two reduced points	yes	split
$p \equiv 3 \pmod{4}$	$\operatorname{Spec} \mathbb{F}_{p^2}$	yes	inert

This example shows how scheme-theoretic fibres encode arithmetic information.

2.3 Examples of Reduced and Nonreduced Fibres

Example 11. A reduced one-point fibre

The scheme

$$\operatorname{Spec} k$$

has one point. Since k is a field, it has no nonzero nilpotent elements.

Therefore

$$\operatorname{Spec} k$$

is reduced and irreducible.

Example 12. Two reduced points

The scheme

$$\operatorname{Spec}(k \times k)$$

has two points.

Indeed, the prime ideals of $k \times k$ are

$$k \times 0 \quad \text{and} \quad 0 \times k.$$

The ring $k \times k$ has no nonzero nilpotent elements, so the scheme is reduced.

However, it is not irreducible, because it is the union of two closed points.

Thus

$$\operatorname{Spec}(k \times k)$$

is reduced but reducible.

Example 13. One nonreduced point

Let

$$R = k[\varepsilon]/(\varepsilon^2).$$

Then

$$\operatorname{Spec} R$$

has one point, because R has one prime ideal,

$$(\varepsilon).$$

But R is not reduced, since

$$\varepsilon \neq 0$$

and

$$\varepsilon^2 = 0.$$

Therefore

$$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$$

is irreducible but not reduced.

This is called the dual number point, or an infinitesimal thickening of a point.

2.4 Examples of Cartesian Diagrams

Example 14. A fibre is a Cartesian square

Let

$$f : X \rightarrow S$$

be a morphism, and let

$$s \in S$$

be a point.

The fibre over s is

$$X_s = X \times_S \operatorname{Spec} \kappa(s).$$

Therefore the diagram

$$\begin{array}{ccc} X_s & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec} \kappa(s) & \longrightarrow & S \end{array}$$

is Cartesian.

Thus every scheme-theoretic fibre is a pullback.

Example 15. Fibre products in sets

In the category of sets, suppose we have maps

$$f : X \rightarrow S, \quad g : Y \rightarrow S.$$

Then the fibre product is

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}.$$

For example, let

$$S = \{0, 1\},$$

$$X = \{a, b, c\}, \quad Y = \{u, v\}.$$

Suppose

$$f(a) = 0, \quad f(b) = 0, \quad f(c) = 1,$$

and

$$g(u) = 0, \quad g(v) = 1.$$

Then

$$X \times_S Y = \{(a, u), (b, u), (c, v)\}.$$

This is the simplest model of a fibre product: pairs whose images in the base agree.

Example 16. Fibre products of affine schemes

Let

$$X = \operatorname{Spec} B, \quad Y = \operatorname{Spec} C, \quad S = \operatorname{Spec} A.$$

Suppose B and C are A -algebras. Then

$$X \times_S Y = \operatorname{Spec}(B \otimes_A C).$$

For example,

$$\mathbb{A}_k^1 \times_{\operatorname{Spec} k} \mathbb{A}_k^1$$

is

$$\operatorname{Spec}(k[x] \otimes_k k[y]).$$

Since

$$k[x] \otimes_k k[y] \cong k[x, y],$$

we get

$$\mathbb{A}_k^1 \times_k \mathbb{A}_k^1 \cong \mathbb{A}_k^2.$$

So the product of two affine lines over k is the affine plane.

Example 17. The diagonal as an equality condition

Let

$$X = \mathbb{A}_k^1 = \operatorname{Spec} k[x].$$

Then

$$X \times_k X = \operatorname{Spec} k[x_1, x_2].$$

The diagonal morphism

$$\Delta : X \rightarrow X \times_k X$$

corresponds to the ring map

$$k[x_1, x_2] \rightarrow k[x]$$

defined by

$$x_1 \mapsto x, \quad x_2 \mapsto x.$$

The image of the diagonal is defined by the equation

$$x_1 = x_2.$$

Equivalently, it is cut out by the ideal

$$(x_1 - x_2).$$

Thus the diagonal imposes equality of two coordinates.

Example 18. The graph as a pullback of the diagonal

Let

$$f : \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^1$$

be defined by

$$f(x) = x^2.$$

The graph of f is the closed subscheme of

$$\mathbb{A}_k^1 \times_k \mathbb{A}_k^1$$

defined by

$$y = x^2.$$

Algebraically, the graph is

$$\Gamma_f = \operatorname{Spec} k[x, y]/(y - x^2).$$

Now the diagonal in

$$\mathbb{A}_k^1 \times_k \mathbb{A}_k^1$$

is defined by

$$y_1 - y_2 = 0.$$

Consider the morphism

$$\mathbb{A}_k^1 \times_k \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^1 \times_k \mathbb{A}_k^1$$

given by

$$(x, y) \mapsto (f(x), y) = (x^2, y).$$

Pulling back the diagonal equation

$$y_1 - y_2 = 0$$

gives

$$x^2 - y = 0.$$

Thus the pullback of the diagonal is exactly the graph

$$y = x^2.$$

This is the concrete affine version of the Cartesian diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \Gamma_f \downarrow & & \downarrow \Delta \\ X \times_S Y & \longrightarrow & Y \times_S Y. \end{array}$$

2.5 Examples Connecting Normalization, Fibres, and Cartesian Diagrams

Example 19. Fibres of a normalization map

Consider the cusp

$$X = \operatorname{Spec} k[x, y]/(y^2 - x^3).$$

Its normalization is

$$\widetilde{X} = \operatorname{Spec} k[t],$$

with normalization map induced by

$$x = t^2, \quad y = t^3.$$

The point $t = 0$ maps to

$$(x, y) = (0, 0),$$

which is the cusp point.

Away from the cusp, the normalization map is essentially one-to-one. Over the cusp point, the fibre consists of one point corresponding to the single branch of the cusp.

For a nodal curve, the normalization map usually has two points over the node, one for each branch.

Thus the fibres of the normalization map describe how singularities are separated.

Example 20. Arithmetic fibres as Cartesian squares

Let

$$X = \operatorname{Spec} \mathbb{Z}[i] = \operatorname{Spec} \mathbb{Z}[T]/(T^2 + 1).$$

For a prime p , the fibre over p is

$$X_p = X \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{F}_p.$$

Therefore there is a Cartesian diagram

$$\begin{array}{ccc} X_p & \longrightarrow & X \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathbb{F}_p & \longrightarrow & \operatorname{Spec} \mathbb{Z}. \end{array}$$

Algebraically,

$$X_p = \operatorname{Spec} \left(\mathbb{Z}[T]/(T^2 + 1) \otimes_{\mathbb{Z}} \mathbb{F}_p \right).$$

Hence

$$X_p = \operatorname{Spec} \mathbb{F}_p[T]/(T^2 + 1).$$

This is exactly the fibre computation from the Gaussian integer example.

Thus Problem 9 is an application of the Cartesian-square construction from Problem 10.

Example 21. Universal properties in all three constructions

The three topics are unified by the following table:

Construction	Object	Universal meaning
Normalization	$\widetilde{X} \rightarrow X$	universal map from normal integral schemes
Fibre	X_s	pullback along $\operatorname{Spec} \kappa(s) \rightarrow S$
Cartesian square	$B \times_D C$	universal object mapping to B and C

Thus the same philosophy appears in all three settings:

algebraic geometry builds spaces by universal constructions.

Normalization is universal among normal models.

Fibres are universal pullbacks to residue fields.

Cartesian diagrams are universal pullback diagrams.

3 Charts, Diagrams, and Tables for Problems 8–10

This section collects visual summaries for Problems 8–10. The main topics are

normalization, scheme-theoretic fibres, Cartesian diagrams.

The common idea is

local algebra + universal properties = geometric constructions.

3.1 Global Overview

Problem	Topic	Construction	Main formula or idea
8	Normalization	$X \rightsquigarrow \widetilde{X}$	$\widetilde{U} = \operatorname{Spec} \overline{A}$
9	Fibres	$f^{-1}(\mathfrak{p})$	$\operatorname{Spec} (B \otimes_A \kappa(\mathfrak{p}))$
10	Cartesian diagrams	pullback square	$A \cong B \times_D C$

Thus the three problems are linked by the following principle:

normalization, fibres, and Cartesian diagrams are all universal constructions.

3.2 Normalization Diagrams and Tables

Normalization pipeline

Let A be an integral domain with fraction field

$$K = \text{Frac}(A).$$

The normalization process is

$$A \subseteq \overline{A} \subseteq K,$$

where $\overline{A}$ is the integral closure of A in K .

Algebraically:

$$A \subseteq \overline{A} \subseteq \text{Frac}(A).$$

Geometrically, the arrows reverse:

$$\text{Spec } \overline{A} \longrightarrow \text{Spec } A.$$

Thus:

$$\begin{array}{ccccc} A & \subseteq & \overline{A} & \subseteq & \text{Frac}(A) \\ \downarrow & & \downarrow & & \\ \text{Spec } A & \longleftarrow & \text{Spec } \overline{A} & & \end{array}$$

The scheme

$$\widetilde{X} = \text{Spec } \overline{A}$$

is the normalization of

$$X = \text{Spec } A.$$

Affine normalization table

Object	Algebraic description	Geometric meaning
X	$\text{Spec } A$	original integral affine scheme
$K(X)$	$\text{Frac}(A)$	function field
$\overline{A}$	integral closure of A in $K(X)$	normal coordinate ring
$\widetilde{X}$	$\text{Spec } \overline{A}$	normalization of X
$\pi : \widetilde{X} \rightarrow X$	$A \hookrightarrow \overline{A}$	normalization morphism

Localization and gluing

The key identity is

$$\overline{A_f} = (\overline{A})_f.$$

More generally, if $S \subseteq A$ is a multiplicative subset, then

$$\overline{S^{-1}A} = S^{-1}\overline{A}.$$

This gives the algebraic diagram

$$\begin{array}{ccc} A & \longrightarrow & A_f \\ \downarrow & & \downarrow \\ \overline{A} & \longrightarrow & (\overline{A})_f = \overline{A_f} \end{array}$$

and the corresponding scheme diagram

$$\begin{array}{ccc} \text{Spec}(\overline{A})_f & \hookrightarrow & \text{Spec } \overline{A} \\ \downarrow & & \downarrow \\ \text{Spec } A_f & \hookrightarrow & \text{Spec } A. \end{array}$$

This explains why affine normalizations glue on overlaps.

Universal property of normalization

If Z is a normal integral scheme and

$$f : Z \rightarrow X$$

is dominant, then f factors uniquely through the normalization

$$\pi : \widetilde{X} \rightarrow X.$$

The diagram is

$$\begin{array}{ccc}
 Z & \xrightarrow{\tilde{f}} & \widetilde{X} \\
 & \searrow f & \swarrow \pi \\
 & X &
 \end{array}$$

Thus

$$f = \pi \circ \tilde{f}.$$

The normalization is therefore the universal normal integral scheme above X .

Cusp normalization table

For the cusp

$$X = \operatorname{Spec} k[x, y]/(y^2 - x^3),$$

we use the parametrization

$$x = t^2, \quad y = t^3.$$

Then

$$k[x, y]/(y^2 - x^3) \cong k[t^2, t^3] \subseteq k[t].$$

Part	Expression	Meaning
Original ring	$A = k[x, y]/(y^2 - x^3)$	coordinate ring of the cusp
Embedded form	$A \cong k[t^2, t^3]$	subring of $k[t]$
Fraction field	$\operatorname{Frac}(A) = k(t)$	rational function field
Missing integral element	t	$t^2 \in A, t^3 \in A, t \notin A$
Integral closure	$\overline{A} = k[t]$	normal ring
Normalization	$\operatorname{Spec} k[t] \rightarrow X$	affine line parametrizes the cusp

The key point is that

$$t$$

is integral over A , because it satisfies

$$T^2 - x = 0,$$

but

$$t \notin A.$$

Therefore A is not normal, and its normalization is $k[t]$.

3.3 Fibre Computation Diagrams and Tables

General fibre formula

Let

$$f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$$

be induced by a ring homomorphism

$$A \rightarrow B.$$

For a point

$$\mathfrak{p} \in \operatorname{Spec} A,$$

the fibre over $\mathfrak{p}$ is

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} (B \otimes_A \kappa(\mathfrak{p})),$$

where

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p})$$

is the residue field of $\mathfrak{p}$.

Thus:

$$\boxed{f^{-1}(\mathfrak{p}) = \operatorname{Spec} (B \otimes_A \kappa(\mathfrak{p}))}.$$

Fibre computation flow

$$\begin{array}{ccccc} A & \longrightarrow & B & & \\ \downarrow & & \downarrow & & \\ \kappa(\mathfrak{p}) & \longrightarrow & B \otimes_A \kappa(\mathfrak{p}) & \longrightarrow & \text{fibre ring} \end{array}$$

Then

$$\boxed{\text{fibre scheme} = \operatorname{Spec}(\text{fibre ring}).}$$

Fibre as a Cartesian square

Every fibre is a pullback.

$$\begin{array}{ccc}
X_{\mathfrak{p}} & \longrightarrow & X \\
\downarrow & & \downarrow f \\
\mathrm{Spec} \kappa(\mathfrak{p}) & \longrightarrow & \mathrm{Spec} A
\end{array}$$

Here

$$X_{\mathfrak{p}} = X \times_{\mathrm{Spec} A} \mathrm{Spec} \kappa(\mathfrak{p}).$$

This is the bridge between Problem 9 and Problem 10.

Fibre computation recipe

Step	Operation	Meaning
1	$\mathfrak{p} \in \mathrm{Spec} A$	choose a point of the base
2	$\kappa(\mathfrak{p}) = \mathrm{Frac}(A/\mathfrak{p})$	compute the residue field
3	$B \otimes_A \kappa(\mathfrak{p})$	specialize the coordinate ring
4	$\mathrm{Spec}(B \otimes_A \kappa(\mathfrak{p}))$	take spectrum of the fibre ring
5	analyse the ring	decide reducedness and irreducibility

Common fibre types

Fibre ring	Geometry	Reduced?	Irreducible?
k	one point	yes	yes
$k \times k$	two reduced points	yes	no
$k[\varepsilon]/(\varepsilon^2)$	doubled point	no	yes
$k[U]/(U^2)$	doubled point	no	yes
$k[U]/((U-a)^2)$	doubled point at a	no	yes
0	empty fibre	yes	usually no

The important lesson is

reduced $\neq$ irreducible.

A doubled point is irreducible but not reduced, while two separated points are reduced but not irreducible.

3.4 Tables for the Fibres in Problem 9

Hyperbola projection

Consider

$$\mathrm{Spec} k[T, U]/(TU - 1) \rightarrow \mathrm{Spec} k[T].$$

The fibre over $T = a$ is

$$\operatorname{Spec} k[U]/(aU - 1).$$

Base point	Fibre ring	Fibre	Type
$a = 0$	0	$\emptyset$	empty
$a \neq 0$	k	$\operatorname{Spec} k$	one reduced point

Geometrically,

$$TU = 1$$

has no point above $T = 0$, and one point above every $T \neq 0$.

Double-cover type fibre

Consider

$$\operatorname{Spec} k[T, U]/(T^2 - U^2) \rightarrow \operatorname{Spec} k[T].$$

The fibre over $T = a$ is

$$\operatorname{Spec} k[U]/(a^2 - U^2).$$

If $\operatorname{char} k \neq 2$, then

$$a^2 - U^2 = (a - U)(a + U).$$

Hence:

Base point	Fibre ring	Reduced?	Irreducible?
$a = 0$	$k[U]/(U^2)$	no	yes
$a \neq 0$	$k[U]/((U - a)(U + a)) \cong k \times k$	yes	no

If $\operatorname{char} k = 2$, then

$$a^2 - U^2 = (U - a)^2.$$

So every fibre is a doubled point:

Base point	Fibre ring	Reduced?	Irreducible?
all a	$k[U]/((U - a)^2)$	no	yes

Plane component and curve component

For the family

$$\mathrm{Spec} k[T, U, V, W] / \left((U + T)W, (U + T)(U^3 + U^2 + UV^2 - V^2) \right) \rightarrow \mathrm{Spec} k[T],$$

the fibre over $T = a$ is defined by

$$(U + a)W = 0,$$

and

$$(U + a)F(U, V) = 0,$$

where

$$F(U, V) = U^3 + U^2 + UV^2 - V^2.$$

Set-theoretically,

$$\text{fibre} = \{U = -a\} \cup \{W = 0, F(U, V) = 0\}.$$

Thus:

$$\begin{aligned} \text{Fibre} &= \text{plane component} \cup \text{curve component} \\ &= \{U = -a\} \cup \{W = 0, F(U, V) = 0\}. \end{aligned}$$

Component	Equations	Geometry
Plane component	$U = -a$	$\mathbb{A}_{\kappa(\mathfrak{p})}^2$ with coordinates V, W
Curve component	$W = 0, F(U, V) = 0$	plane curve

Reducedness summary:

Characteristic	Reduced?	Irreducible?
$\text{char } k \neq 2$	yes	no
$\text{char } k = 2$	no	no

Affine line over $\mathrm{Spec} \mathbb{Z}$

Consider

$$\mathrm{Spec} \mathbb{Z}[T] \rightarrow \mathrm{Spec} \mathbb{Z}.$$

Point of $\mathrm{Spec} \mathbb{Z}$	Residue field	Fibre	Meaning
(0)	$\mathbb{Q}$	$\mathrm{Spec} \mathbb{Q}[T]$	$\mathbb{A}_{\mathbb{Q}}^1$
(p)	$\mathbb{F}_p$	$\mathrm{Spec} \mathbb{F}_p[T]$	$\mathbb{A}_{\mathbb{F}_p}^1$

So

$$\mathrm{Spec} \mathbb{Z}[T]$$

is a family of affine lines over all prime fields, together with the generic fibre over $\mathbb{Q}$.

Gaussian integers

Consider

$$\mathrm{Spec} \mathbb{Z}[i] = \mathrm{Spec} \mathbb{Z}[T]/(T^2 + 1) \rightarrow \mathrm{Spec} \mathbb{Z}.$$

The fibre over p is

$$\mathrm{Spec} \mathbb{F}_p[T]/(T^2 + 1).$$

Prime	Factorization mod p	Fibre	Reduced?	Arithmetic meaning
2	$T^2 + 1 = (T + 1)^2$	doubled point	no	ramified
$p \equiv 1 \pmod{4}$	$T^2 + 1 = (T - a)(T + a)$	two points	yes	split
$p \equiv 3 \pmod{4}$	$T^2 + 1$ irreducible	$\mathrm{Spec} \mathbb{F}_{p^2}$	yes	inert

This table shows how arithmetic behaviour is encoded by scheme-theoretic fibres.

The morphism $\mathrm{Spec} \mathbb{C} \rightarrow \mathrm{Spec} \mathbb{Z}$

The ring map is

$$\mathbb{Z} \rightarrow \mathbb{C}.$$

The fibre over the generic point is

$$\mathrm{Spec} (\mathbb{C} \otimes_{\mathbb{Z}} \mathbb{Q}) \cong \mathrm{Spec} \mathbb{C}.$$

The fibre over (p) is

$$\mathrm{Spec} (\mathbb{C} \otimes_{\mathbb{Z}} \mathbb{F}_p) \cong \mathrm{Spec} (\mathbb{C}/p\mathbb{C}).$$

Since p is invertible in $\mathbb{C}$,

$$p\mathbb{C} = \mathbb{C}.$$

Therefore

$$\mathbb{C}/p\mathbb{C} = 0.$$

Thus:

Point of $\operatorname{Spec} \mathbb{Z}$	Fibre ring	Fibre	Reason
(0)	$\mathbb{C} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{C}$	$\operatorname{Spec} \mathbb{C}$	$\mathbb{C}$ is a $\mathbb{Q}$ -algebra
(p)	$\mathbb{C} \otimes_{\mathbb{Z}} \mathbb{F}_p \cong \mathbb{C}/p\mathbb{C} = 0$	$\emptyset$	p is invertible in $\mathbb{C}$

So the morphism

$$\operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{Z}$$

hits only the generic point.

3.5 Cartesian Diagram Charts

General Cartesian square

A commutative square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

is Cartesian if

$$A \cong B \times_D C.$$

Equivalently, for every object Q , maps

$$Q \rightarrow A$$

are the same as pairs of maps

$$Q \rightarrow B, \quad Q \rightarrow C,$$

whose compositions to D agree.

Thus:

$$\operatorname{Hom}(Q, A) \cong \operatorname{Hom}(Q, B) \times_{\operatorname{Hom}(Q, D)} \operatorname{Hom}(Q, C).$$

Map into A	Equivalent data
$Q \rightarrow A$	$Q \rightarrow B, \quad Q \rightarrow C, \quad \text{with equal compositions to } D$

Fibre product of sets

If

$$f : X \rightarrow S, \quad g : Y \rightarrow S,$$

then

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}.$$

The diagram is

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & Y \\ \downarrow & & \downarrow g \\ X & \xrightarrow{f} & S. \end{array}$$

The interpretation is

fibre product = pairs with the same image in the base.

Fibre product of affine schemes

Let

$$X = \operatorname{Spec} B, \quad Y = \operatorname{Spec} C, \quad S = \operatorname{Spec} A.$$

If B and C are A -algebras, then

$$X \times_S Y = \operatorname{Spec}(B \otimes_A C).$$

The Cartesian square is

$$\begin{array}{ccc} \operatorname{Spec}(B \otimes_A C) & \longrightarrow & \operatorname{Spec} C \\ \downarrow & & \downarrow \\ \operatorname{Spec} B & \longrightarrow & \operatorname{Spec} A. \end{array}$$

Thus:

$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C = \operatorname{Spec}(B \otimes_A C).$

3.6 Diagrams for Problem 10

First Cartesian diagram: diagonal imposes equality over S

Suppose we have morphisms

$$X \rightarrow S \rightarrow T, \quad Y \rightarrow S \rightarrow T.$$

The diagram is

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & S \\ \downarrow & & \downarrow \Delta \\ X \times_T Y & \longrightarrow & S \times_T S. \end{array}$$

The object

$$X \times_T Y$$

only requires equality after mapping to T .

But pulling back along

$$\Delta : S \rightarrow S \times_T S$$

forces the two maps to S to be equal.

Therefore

$$X \times_S Y \cong (X \times_T Y) \times_{S \times_T S} S.$$

The meaning can be summarized as follows:

Object	Condition
$X \times_T Y$	x and y have the same image in T
$S \times_T S$	pairs (s_1, s_2) with the same image in T
$\Delta(S)$	$s_1 = s_2$
$X \times_S Y$	x and y have the same image in S

So the diagonal changes the condition

equal over T

into the stronger condition

equal over S .

Second Cartesian diagram: graph as pullback of diagonal

Let

$$f : X \rightarrow Y$$

be a morphism over S .

The graph of f is

$$\Gamma_f : X \rightarrow X \times_S Y.$$

The diagonal of Y is

$$\Delta : Y \rightarrow Y \times_S Y.$$

The diagram is

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \Gamma_f \downarrow & & \downarrow \Delta \\ X \times_S Y & \longrightarrow & Y \times_S Y. \end{array}$$

The bottom map sends

$$(x, y) \mapsto (f(x), y).$$

Pulling back along the diagonal imposes the condition

$$(f(x), y) \in \Delta(Y).$$

But this means

$$f(x) = y.$$

Therefore the pullback is

$$\{(x, y) : y = f(x)\},$$

which is exactly the graph of f .

The table is:

Object	Meaning
$X \times_S Y$	pairs (x, y) compatible over S
$Y \times_S Y$	pairs (y_1, y_2) compatible over S
$\Delta(Y)$	pairs (y, y)
$\Gamma_f(X)$	pairs $(x, f(x))$

Hence

graph of f = pullback of the diagonal of Y .

3.7 Final Visual Summary

Construction	Algebra	Geometry	Diagram type
Normalization	$A \subseteq \overline{A}$	$\widetilde{X} \rightarrow X$	universal factorization
Fibre	$B \otimes_A \kappa(\mathfrak{p})$	$X_{\mathfrak{p}}$	Cartesian square
Fibre product	$B \otimes_A C$	$X \times_S Y$	Cartesian square
Diagonal	$x_1 = x_2$	$\Delta : X \rightarrow X \times_S X$	equality condition
Graph	$y = f(x)$	$\Gamma_f : X \rightarrow X \times_S Y$	pullback of diagonal

One-page memory diagram

Normalization	$A \subseteq \overline{A}$	$\implies$	$\mathrm{Spec} \overline{A} \rightarrow \mathrm{Spec} A$	normal model
Fibre	$A \rightarrow B, \mathfrak{p} \in \mathrm{Spec} A$	$\implies$	$\mathrm{Spec}(B \otimes_A \kappa(\mathfrak{p}))$	specialization
Cartesian square	$A \cong B \times_D C$	$\implies$	universal pullback	compatibility condition

The final slogan is

Normalization repairs spaces, fibres inspect families, and Cartesian diagrams organize pullbacks.